

Simon Says on Naoqi Robot

RoboCup

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Overview

Implementation of a Simon Says game using the robot
Combines multiple core capabilities:

- **Speech:** speech recognition and speech output
- **Vision:** recognition of objects
- **Motion:** execution of robot movements

Modular design:

- Easily extendable with new commands, gestures or rules
- Solid basis for further human-robot interaction experiments



Figure 1: Naoqi Robot [1].

Implementation

Note: Machine Learning runs in [Python 3.13.0](#), NAOqi communication runs in [Python 2.7](#) inside Docker.

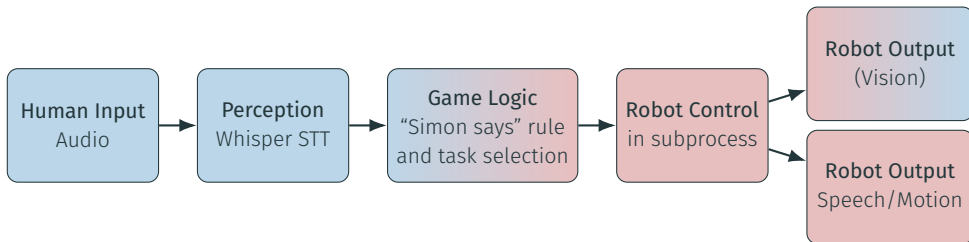


Figure 2: Processing pipeline of the Simon Says game.

“Simon says: Raise left arm.”

“Simon says: Capture frame”

“Solve the this Integral $\int \frac{1}{x} dx$.”

“Simon says: Move Forward”

“Simon says: Change eye color to red.”

Demonstration

“Simon says: Turn head.”

“Simon says: Sit”

“Simon says: Stand”

“Do a backflip”

“Do a head spin.”

References



Aldebaran Robotics.

Naoqi documentation.

aldebaran.com, 2017.

Accessed: Jan 2026.



Ramin Leon Neymeyer, Ramon Rennert, Anton Dilg, and Georg Wolnik.

Nao says robot game.

github.com/lar-tech/nao-says-robot-game, 2026.